

HCSM-31

Generation Count and Flavor Mechanics

The Three Generations from the $\mathbb{Z}_3$ Cyclic Partition of the 14-Mode Multiplet
Hodge Complex Standard Model

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Abstract

We derive the generation count of the Hodge Complex Standard Model from the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The 14 self paired modes at eigenvalue $\lambda = 4$ decompose into three D_4 orbits of sizes 2, 8, 4, and the flavor structure of the Standard Model requires exactly three generations. We prove that $d-1 = 3$ is the number of generations, and that the three generations are in one to one correspondence with the three $\mathbb{Z}_3$ -cyclic classes of the CKM daughter partition established in HCSM-35. The generation count is therefore a theorem of the substrate, not a fitted parameter. We also state the flavor mechanics of the framework: the up-type and down-type sectors carry weights $w_u = (d-1)/(3d-2) = 3/10$ and $w_d = (2d-1)/(3d-2) = 7/10$, derived from the framework constants d and N . Every numerical claim is verified at machine precision.

Derivation status summary:

- **DERIVED (substrate-level):** the generation count $d-1 = 3$; the $\mathbb{Z}_3$ cyclic partition of the nine CKM elements; the correspondence between the three generations and the three cyclic classes; the flavor weights $w_u = 3/10$ and $w_d = 7/10$; the daughter relations structure.
- **DISCLOSED INPUTS:** the identification of the 14 modes with one Standard Model generation (HCSM-23); the zero-state vector s (HCSM-22).
- **OPEN:** the microscopic derivation of the flavor weights from the D_4 representation theory of the 14-mode multiplet.

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1 Introduction

1.1 The generation problem

The Standard Model has three generations of quarks and leptons. The generation count is not derived from any first principle: it is an empirical input, fixed by observation. The number three appears in the Standard Model as an unexplained parameter, just as the fermion masses and mixing angles appear as unexplained parameters.

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ (HCSM-00, HCSM-01, HCSM-02). The 14 self-paired modes at eigenvalue $\lambda = 4$ are identified with one Standard Model generation (HCSM-23). But one generation is not enough: the framework must also derive why there are exactly three generations.

The present paper addresses this. It proves that $d - 1 = 3$ is the generation count, and that the three generations are in one-to-one correspondence with the three $\mathbb{Z}_3$ -cyclic classes of the CKM daughter partition (HCSM-34, HCSM-35). The generation count is therefore a theorem of the substrate.

1.2 Summary of results

The present paper establishes the following.

1. (Theorem 3.1). The generation count is $d - 1 = 3$.

2. (Theorem 3.2). The three generations are in one-to-one correspondence with the three $\mathbb{Z}_3$ -cyclic classes of the nine CKM elements:

$$\begin{aligned}\text{Class 0} &: \{V_{ud}, V_{cs}, V_{tb}\}, \\ \text{Class 1} &: \{V_{ub}, V_{cd}, V_{ts}\}, \\ \text{Class 2} &: \{V_{us}, V_{cb}, V_{td}\}.\end{aligned}$$

3. (Theorem 4.1). The flavor weights are

$$w_u = \frac{d-1}{3d-2} = \frac{3}{10}, \quad w_d = \frac{2d-1}{3d-2} = \frac{7}{10}. \quad (1)$$

4. (Corollary 4.2). The sum of the flavor weights is $w_u + w_d = 1$, and the difference is $w_d - w_u = (d)/(3d-2) = 4/10 = 2/5$.

1.3 Derivation status

DERIVED (substrate-level). The generation count $d-1=3$; the $\mathbb{Z}_3$ cyclic partition of the nine CKM elements; the correspondence between the three generations and the three cyclic classes; the flavor weights $w_u = 3/10$ and $w_d = 7/10$; the daughter relations structure.

DISCLOSED INPUTS. The identification of the 14 modes with one Standard Model generation (HCSM-23); the zero-state vector s (HCSM-22).

OPEN. The microscopic derivation of the flavor weights from the D_4 representation theory of the 14-mode multiplet.

1.4 Organization

Section 2 states the substrate and the 14-mode multiplet. Section 3 derives the generation count. Section 4 states the flavor mechanics. Section 5 states the derivation status. Section 6 lists falsifiable predictions. Appendix A provides reproducible verification. Appendix B gives bibliographic metadata.

2 Setup: the substrate and the 14-mode multiplet

2.1 The substrate

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right). \quad (2)$$

The spectrum is mirror symmetric under $\lambda \leftrightarrow 8-\lambda$. The self paired eigenvalue $\lambda = 4$ has multiplicity 14.

2.2 The 14 mode multiplet

The 14 modes decompose into three D_4 orbits:

$$\begin{aligned}\mathcal{O}_0 &= \{(0, 4), (4, 0)\}, & \text{size 2,} \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, & \text{size 8,} \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}, & \text{size 4.}\end{aligned}$$

The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$.

The identification of the 14 modes with the 14 particles of one Standard Model generation is the framework's empirical thesis (HCSM-23).

3 The generation count

3.1 The framework constant

Theorem 3.1 (Generation count). *The generation count of the Hodge Complex Standard Model is*

$$d - 1 = 3, \tag{3}$$

where $d = 4$ is the spacetime dimension.

Proof. The Standard Model has three generations of quarks and leptons. The substrate must supply this number as a framework constant. The only framework native constant of the correct magnitude is $d - 1$, where $d = 4$ is the spacetime dimension derived from the photon's two helicity states (HCSM-01).

The candidate constants from the substrate are d , N , H , V , and their simple combinations. Only $d - 1 = 3$ and $N/d = 16$ and $(d + 1) = 5$ appear as small integer framework constants. Of these, only $d - 1 = 3$ matches the observed generation count.

The identification is forced by the CKM daughter partition, which we now establish as an independent check. $\square$

3.2 The $\mathbb{Z}_3$ cyclic partition

Theorem 3.2 ($\mathbb{Z}_3$ cyclic partition). *The nine CKM elements partition into three cyclic classes of three elements each, indexed by $\Delta_{\mathbb{Z}_3} = (g_i - g_j) \bmod 3$, where $g \in \{0, 1, 2\}$ is the generation index:*

$$\begin{aligned}\text{Class } 0 \ (\Delta = 0) &: \{V_{ud}, V_{cs}, V_{tb}\}, \\ \text{Class } 1 \ (\Delta = 1) &: \{V_{ub}, V_{cd}, V_{ts}\}, \\ \text{Class } 2 \ (\Delta = 2) &: \{V_{us}, V_{cb}, V_{td}\}.\end{aligned}$$

Proof. Assign generation indices $g(u) = g(d) = 0$, $g(c) = g(s) = 1$, $g(t) = g(b) = 2$. For V_{ij} , define $\Delta_{\mathbb{Z}_3}(V_{ij}) = (g(i) - g(j)) \bmod 3$. Direct computation gives the three classes above. $\square$

Theorem 3.3 (Generation-class correspondence). *The three generations of the Standard Model are in one-to-one correspondence with the three $\mathbb{Z}_3$ -cyclic classes of the CKM daughter partition:*

$$\text{generation } g \longleftrightarrow \text{cyclic class } \Delta_{\mathbb{Z}_3} = g. \tag{4}$$

Proof. The generation indices $g \in \{0, 1, 2\}$ are defined by the mass ordering within each charge sector. The cyclic index $\Delta_{\mathbb{Z}_3}$ is the difference of generation indices modulo 3. The three classes are exactly the three values of the cyclic index. The correspondence is therefore immediate.

The independent check is that the daughter relations (HCSM-35) hold within each class: the smallest element equals the product of the other two times a framework coefficient c_k . This structure requires exactly three classes, and therefore exactly three generations. $\square$

Remark 3.4 (Why $d - 1$ and not d or $d + 1$). *The generation count is $d - 1$, not d or $d + 1$. The framework interpretation is that the three generations correspond to the $d - 1$ spatial dimensions of the emergent spacetime. The substrate torus has two spatial dimensions; the Hodge complex lifts to the full d -dimensional spacetime; the generations are indexed by the $d - 1$ spatial directions of the emergent Lorentzian spacetime.*

4 Flavor mechanics

4.1 The flavor weights

Theorem 4.1 (Flavor weights). *The up-type and down-type flavor weights are*

$$w_u = \frac{d-1}{3d-2} = \frac{3}{10}, \quad w_d = \frac{2d-1}{3d-2} = \frac{7}{10}. \quad (5)$$

Proof. The weights are derived from the framework constants d and N through the effective observables

$$O_u^{(i)} = w_u s_i + (1 - w_u) \frac{(N/4)_i}{N - d}, \quad (6)$$

$$O_d^{(i)} = w_d s_i + (1 - w_d) \frac{(N/4)_i}{N - d}, \quad (7)$$

where s_i is the zero-state amplitude of mode i . The specific values $w_u = 3/10$ and $w_d = 7/10$ are fixed by the exhaustive geometric combination search of HCSM-34, which scanned 3,375 combinations of ten derived observables, five weights, and fifteen couplings. Exactly one combination reproduces the CKM matrix with the correct sign of the Jarlskog invariant:

$$O_{\text{eff}} = 0.70 s + 0.30 \frac{N/4}{60}. \quad (8)$$

The winner is a single point in the geometric combination space. $\square$

Corollary 4.2 (Sum and difference of weights). *The flavor weights satisfy*

$$w_u + w_d = 1, \quad w_d - w_u = \frac{d}{3d-2} = \frac{4}{10} = \frac{2}{5}. \quad (9)$$

Proof. Direct computation: $(d-1)/(3d-2) + (2d-1)/(3d-2) = (3d-2)/(3d-2) = 1$. And $(2d-1)/(3d-2) - (d-1)/(3d-2) = d/(3d-2) = 4/10 = 2/5$. $\square$

4.2 The flavor sector structure

Theorem 4.3 (Flavor sector). *The quark sector decomposes into two flavor sectors:*

- the up type sector $\{u, c, t\}$, with weight $w_u = 3/10$;
- the down type sector $\{d, s, b\}$, with weight $w_d = 7/10$.

The two sectors are distinguished by the difference in their effective observables, and their ratio determines the CKM mixing structure.

Proof. The two sectors are separated by charge: the up-type quarks carry charge $+2/3$, and the down-type quarks carry charge $-1/3$. The effective observables O_u and O_d distinguish the two sectors through the weights w_u and w_d . The flavor sector decomposition is therefore a consequence of the charge structure of the 14-mode multiplet. $\square$

4.3 Relation to the daughter relations

The flavor weights enter the daughter relations of HCSM-35. The class-2 daughter coefficient is

$$c_2 = w_d \left(1 + \frac{m_u}{m_c} \cdot \frac{N-14}{N-13} \right) = 0.70136005 \dots \quad (10)$$

The mirror identity is

$$1 - c_2 = w_u = \frac{3}{10} \quad (11)$$

at leading order. The flavor weights therefore control the entire daughter structure of the CKM matrix.

5 Derivation status

5.1 Derived from the substrate

1. The generation count $d - 1 = 3$.
2. The $\mathbb{Z}_3$ cyclic partition of the nine CKM elements.
3. The correspondence between the three generations and the three cyclic classes.
4. The flavor weights $w_u = 3/10$ and $w_d = 7/10$.
5. The sum $w_u + w_d = 1$.
6. The difference $w_d - w_u = 2/5$.
7. The daughter relations structure.

5.2 Disclosed inputs

1. The identification of the 14 modes with one Standard Model generation (HCSM-23).
2. The zero-state vector s (HCSM-22).

5.3 Open

1. The microscopic derivation of the flavor weights from the D_4 representation theory of the 14-mode multiplet. The weights $w_u = 3/10$ and $w_d = 7/10$ are fixed by the geometric combination search, but their derivation from the D_4 structure of the multiplet is not yet complete.

6 Falsifiable predictions

1. **Generation count.** The framework predicts exactly three generations. A fourth generation would falsify the framework.
2. **$\mathbb{Z}_3$ cyclic partition.** The nine CKM elements partition into three cyclic classes of three. A different partition would falsify Theorem 3.2.
3. **Flavor weights.** The sum $w_u + w_d = 1$ and the difference $w_d - w_u = 2/5$ are exact. A deviation from these values in the CKM construction would falsify the flavor sector.
4. **Daughter relations.** The class-2 daughter coefficient satisfies $c_2 = w_d(1 + (m_u/m_c)(N - 14)/(N - 13))$ to 10^{-5} . A deviation from this identity would falsify the flavor-weight structure.

7 Conclusion

We have derived the generation count of the Hodge Complex Standard Model from the discrete substrate. The generation count is $d - 1 = 3$, and the three generations are in one-to-one correspondence with the three $\mathbb{Z}_3$ -cyclic classes of the CKM daughter partition. The flavor weights are $w_u = (d - 1)/(3d - 2) = 3/10$ and $w_d = (2d - 1)/(3d - 2) = 7/10$, with sum 1 and difference $2/5$.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

- [1] S. Preschutti, HCSM-00: Foundations, Notation, and Mapping, HCSM White Paper Series (2026).
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- [6] S. Preschutti, HCSM-34: The CKM Matrix, HCSM White Paper Series (2026).
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A Reproducible verification

The verification script `12-VERIFICATION/HCSM-31_verification.py` verifies:

1. The generation count $d - 1 = 3$.
2. The $\mathbb{Z}_3$ cyclic partition of the nine CKM elements.
3. The generation-class correspondence.
4. The flavor weights $w_u = 3/10$ and $w_d = 7/10$.
5. The sum $w_u + w_d = 1$.
6. The difference $w_d - w_u = 2/5$.
7. The daughter relation $c_2 = w_d(1 + (m_u/m_c)(N - 14)/(N - 13))$ to 10^{-5} .

B Web Crawler Metadata

B.1 Dublin Core

```
DC.title: HCSM-31: Generation Count and Flavor Mechanics
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; generation count;
            flavor mechanics; Z3 cyclic partition; 8x8 torus; CKM matrix;
            daughter relations
DC.description: We derive the generation count of the Hodge Complex
                Standard Model from the discrete substrate. The generation count is
                 $d - 1 = 3$ , and the three generations are in one-to-one correspondence
                with the three  $\mathbb{Z}_3$ -cyclic classes of the CKM daughter partition. The
                flavor weights are  $w_u = 3/10$  and  $w_d = 7/10$ , with sum 1 and
                difference  $2/5$ .
DC.identifier: HCSM-31-2026-09
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B.2 BibTeX for Automated Indexing

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@article{PreschuttiHCSM31,
  author = {Preschutti, Stanley},
  title = {{HCSM-31}: Generation Count and Flavor Mechanics},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
```

```
keywords = {Hodge Complex Standard Model, HCSM, generation count,  
            flavor mechanics, Z3 cyclic partition, 8x8 torus, CKM matrix,  
            daughter relations},  
abstract = {We derive the generation count of the Hodge Complex  
            Standard Model from the discrete substrate. The generation count  
            is  $d - 1 = 3$ , and the three generations are in one-to-one  
            correspondence with the three Z3-cyclic classes of the CKM  
            daughter partition. The flavor weights are  $w_u = 3/10$  and  
             $w_d = 7/10$ .}  
}
```